Introductory Economics for Business and Finance (IC103)

Tutorial 3



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Where business comes to life

Question 1

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In indifference curve analysis, the consumer will maximise utility at the point where:

a) The budget line intersects the indifference curve
b) The budget line is tangential to the indifference curve
c) The budget line is above the indifference curve
d) The budget line is below the indifference curve

Question 1

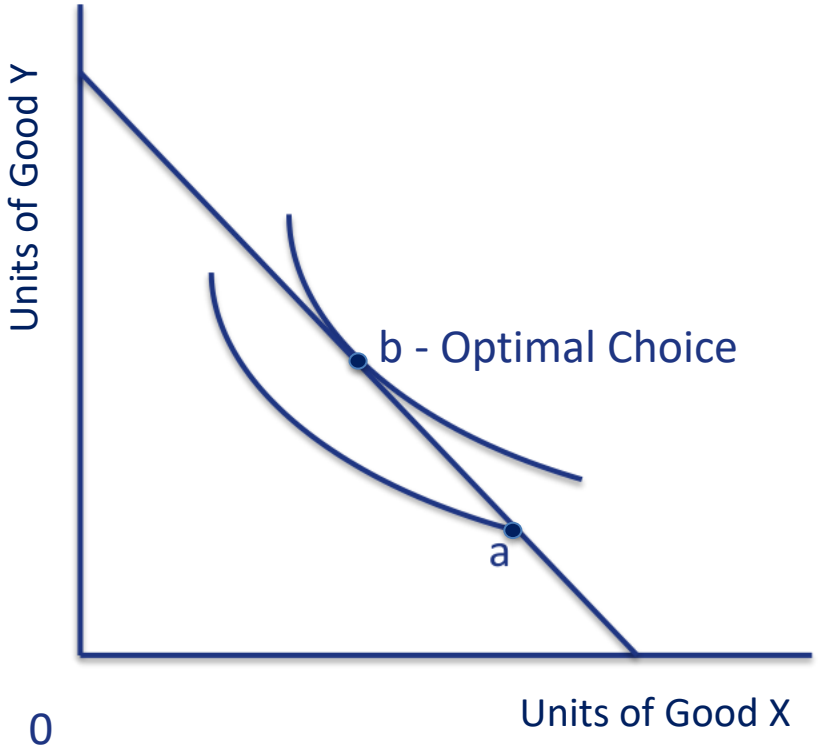
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An indifference curve shows combinations of two goods that give the consumer the same level of satisfaction. A consumer maximises utility by choosing the highest indifference curve they can afford. Usually, this occurs where the budget line is tangent to an indifference curve.

Question 1



Question 2

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When the price of a good changes, a consumer will respond by changing the quantity demanded. The part of the change that is called the substitution effect represents the consumer's response to:

(a) The change in the consumer's nominal income

(b) The change in the consumer's real income

(c) The change in relative prices

(d) An induced change in the consumer's preferences

Question 2

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(b) The change in the consumer's real income

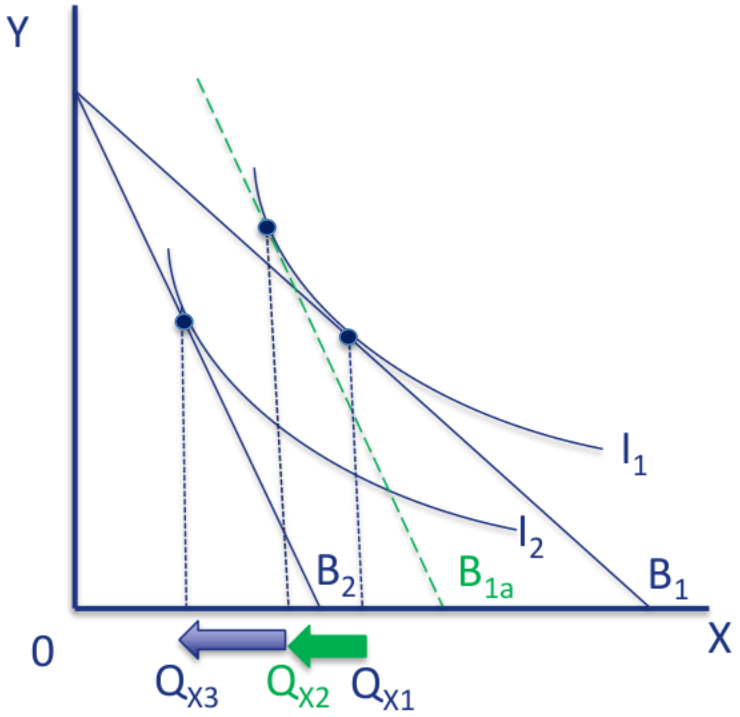
(c) The change in relative prices

(d) An induced change in the consumer's preferences

The substitution effect happens when consumers change their consumption because relative prices change. For example, if the price of coffee increases while the price of tea stays the same, consumers may buy less coffee and more tea.
The income effect happens because a price change also affects consumers' purchasing power.

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Normal good

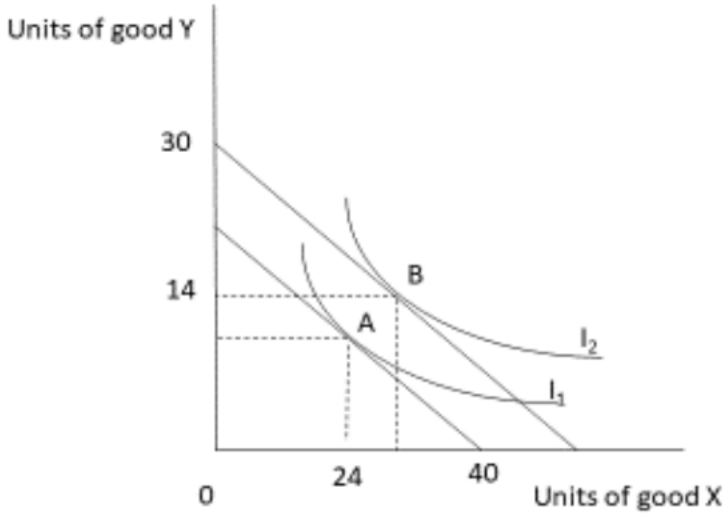


The income and substitution effect both lead to a reduction in consumption of the good whose price has risen

Question 3

Assume that the price of X is £20 and the price of Y is £40. If indifference curve I_2 represents the highest level of utility the consumer can achieve, what is their income?

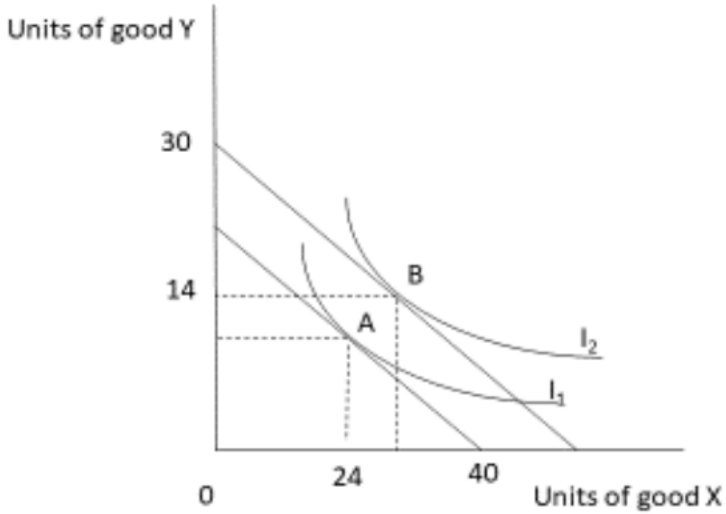
a) £480
b) £600
c) £800
d) £1200



Question 3

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The consumer's income can be calculated using the budget constraint: $Income = P_X X + P_Y Y$. The maximum quantity of Y is 30 units. If the consumer spends all their income on Y: $Income = 0 + £40 * 30 = £1{,}200$.

Question 4

Referring to the diagram in question (3), if the consumer locates at point B, how many units of X will they consume?

a) 28

b) 30

c) 32

d) 60

Question 4

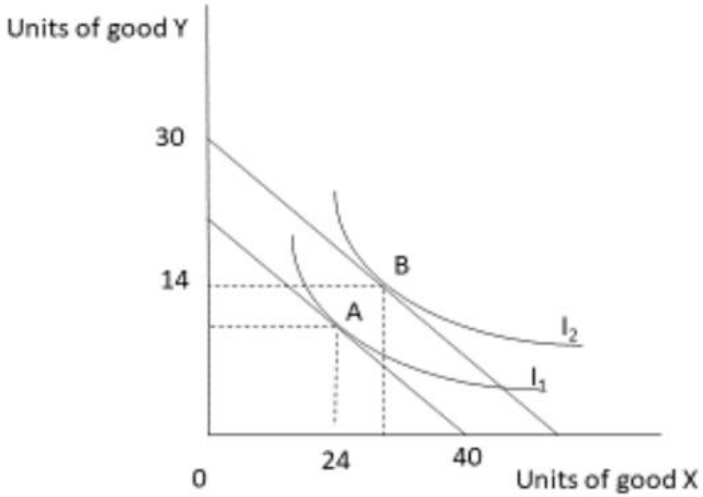
Referring to the diagram in question (3), if the consumer locates at point B, how many units of X will they consume?

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At point B, the consumer buys 14 units of Y.

Cost of Y = 14 × £40 = £560.

Money remaining for X = £1,200 − £560 = £640.

Quantity of X = £640 / £20 = 32 units.

Question 5

When deciding between an expensive, high-quality product and a cheaper, lower-quality product, which of the following offers the best value for money?

(a) The product with the lowest price
(b) The product with the highest quality
(c) The product that provides the highest total utility
(d) The product that provides the highest utility per pound spent

Question 5

When deciding between an expensive, high-quality product and a cheaper, lower-quality product, which of the following offers the best value for money?

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(c) The product that provides the highest total utility
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Question 5

When making consumption decisions, consumers consider both the satisfaction they receive and the price they pay. A product offering greater utility per pound spent provides better value for money.

The equi-marginal principle states that consumers maximise utility when the marginal utility per pound spent is equal across goods:

$$\frac{MU_A}{P_A} = \frac{MU_B}{P_B}$$

Here, MU represents marginal utility (additional satisfaction), and P represents price.

If one good provides greater marginal utility per pound, consumers can increase total satisfaction by spending more on that good and less on another.

Question 6

Steve spends his income of £120 on only concerts and cinema tickets. The concerts cost £15 per ticket and the cinema costs £10 per ticket.

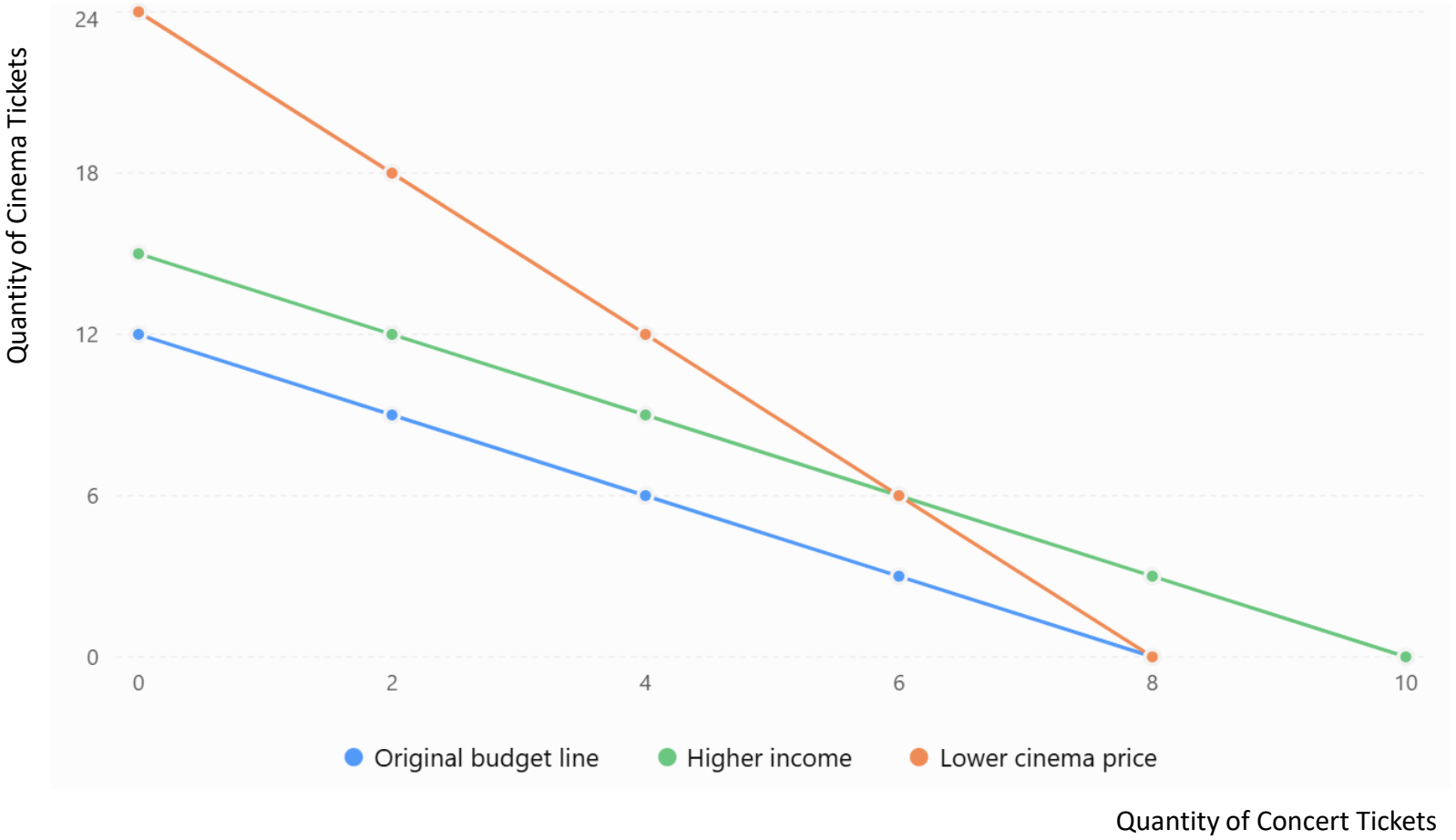
a) Calculate and plot his budget constraint.
b) What happens if his income increases to £150?
c) What happens if his income stays at £120 but cinema tickets fall in price to £5?

Question 6

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	Original (a)	Income increase (b)	Cinema price decrease (c)
Income	£120	£150	£120
Concert price	£15	£15	£15
Cinema price	£10	£10	£5
Max. concerts	£120/£15 = 8	10	8
Max. cinema tickets	£120/£10 = 12	15	24
Budget-line change	Original	Parallel outward shift	Outward rotation

Question 6


Quantity of Cinema Tickets
24
18
12
6
0
0
2
4
6
8
10
Original budget line
Higher income
Lower cinema price
Quantity of Concert Tickets

Question 7

Max employs labour and capital to produce widgets. Currently, the last unit of labour employed has a marginal product of 100 units. The last unit of capital employed has a marginal product of 40 units. The price of labour is £25 per unit and the price of capital is £10 per unit. Max should:

(a) Use more of both factors of production

(b) Hire more labour and use less capital

(c) Hire less labour and more capital

(d) Keep the current combination of labour and capital, as both factors provide the same marginal product per pound spent

Question 7

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(c) Hire less labour and more capital

(d) Keep the current combination of labour and capital, as both factors provide the same marginal product per pound spent

Question 7

Max uses two inputs of production to produce widgets:
Labour: Marginal product = 100 units; Price = £25
Capital: Marginal product = 40 units; Price = £10

To produce a given output at the lowest cost, the firm should use labour and capital so that the marginal product per pound spent is equal for both inputs.

Apply the equi-marginal principle: $\frac{MP_L}{P_L} = \frac{MP_K}{P_K}$

For labour: 100/25 = 4
Every £1 spent on labour produces 4 additional units of output.
For capital: 40/10 = 4
Every £1 spent on capital also produces 4 additional units of output.

Since both factors produce the same additional output per pound spent, Max has no reason to change the combination of labour and capital.

Question 8

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If a firm is producing at a point where diminishing marginal returns have set in, which of these statement(s) is/are true?

(i) Each additional unit of output will be more expensive to produce

(ii) Each additional unit of output will require increasing amounts of the variable factor of production

(iii) The marginal product of the variable factor of production decreases as the quantity used increases

(a) (i) only

(b) (i) and (ii)

(c) (ii) and (iii)

(d) All are true

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(iii) The marginal product of the variable factor of production decreases as the quantity used increases

Correct answer: (d) All are true

Question 8

Diminishing marginal returns occur when adding more units of a variable factor of production, such as labour, leads to progressively smaller increases in output, while other factors remain fixed.

As a result:

- Each additional worker produces less extra output.
- More labour is needed to produce each additional unit of output.
- The marginal cost of producing additional output increases, assuming the cost per worker stays the same.

Question 8

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For example, the second worker increases total output by 15 units, while the third worker increases it by only 10 units. Diminishing marginal returns begin with the third worker.

Workers	Output	Marginal product
1	10	10
2	25	15
3	35	10
4	43	8
5	48	5
6	51	3
7	52	1

